

Beyond Pip Counting: Interpretable Models for Catan Opening Placement

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Predicting

- Opening settlement placement is one of the most important decisions in **Settlers of Catan**, though players often rely on simple heuristics such as **pip counting**, which places settlements based on the hexes most likely to produce resources.
- We investigate whether a learned model can beat pip counting at predicting wins, and whether the model's learned signals are **interpretable**.
- The input is a single player's opening settlements and roads, the board layout, and turn order; the output is the probability the player wins and a predicted final victory-point margin.
- Predicting a win on just pip counting (AUC 0.54) is nearly no better than random, while a linear model with engineered features reaches AUC 0.72.**

Data

- Using **Catanatron's strongest bot, AlphaBetaPlayer**, we generated **99,999** four-player self-play games (399,996 player-openings) with official Catan spiral number placement and randomized hex and port layouts.
- The label columns are boolean **won** and **vp_margin** (final VP minus max opponent VP).
- We used a 70/15/15 train/val/test split at the game level, so all four rows of a given game land in the same partition.

Features

We engineer **52 features across 8 groups**; raw inputs are hex pip values and resource types adjacent to each settlement.

- Pip score** — summed dice roll probability per settlement
- Resource diversity** — distinct adjacent resources
- Port information** — on-port boolean, BFS distance to nearest port
- Expansion potential** — pip-sum of future legal settlement spots
- Opponent information** — min/mean/max opponent pip, resource overlap
- Resource scarcity** — each settlement's share of total pip per resource
- Engines** — ore+wheat for cities, wood+brick for roads
- Seat index** — placement order 1–4, snake draft rules

Models

We fit five model families for both win classification and VP-margin regression:

- Logistic Regression** — $P(\text{won} = 1 \mid x) = \sigma(\theta^\top x)$, ℓ_2 -regularized cross-entropy
- Ridge Regression** — minimizes $\|y - X\theta\|_2^2 + \alpha\|\theta\|_2^2$, $\alpha = 1$
- XGBoost** — K shallow trees, depth 5, learning rate 0.1, early stopping at 300 trees
- MLP** — two hidden layers (128, 64), ReLU activations, $\ell_2 = 10^{-3}$
- SVM** — linear SVC/SVR ($C = 1$) and RBF-kernel on $\sim 32\text{K}$ subsample

For interpretation we compute **SHAP** values on the XGBoost model and distill it into shallow decision trees (depth 2–4) to produce human-readable if-then rules. For **unsupervised** structure, we run **k -means** ($k = 5$) on standardized opening features without win labels.

Results

Learned models substantially beat pip counting; the table below reports held-out test performance, with training set size 279,597 and test set size 59,998 player-openings.

Model	AUC	Brier	RMSE	R^2
Pip-only baseline	0.536	0.187	3.785	0.004
Logistic / Ridge	0.721	0.166	3.440	0.178
LinearSVC / LinearSVR	0.721	0.166	3.517	0.140
SVM (RBF, $\sim 32\text{K}$)	0.679	0.173	3.502	0.148
MLP	0.732	0.164	3.388	0.202
XGBoost	0.733	0.164	3.384	0.204

Table 1. Lower Brier/RMSE and higher AUC/ R^2 are better. Cross-validation intervals: ± 0.002 – 0.003 .

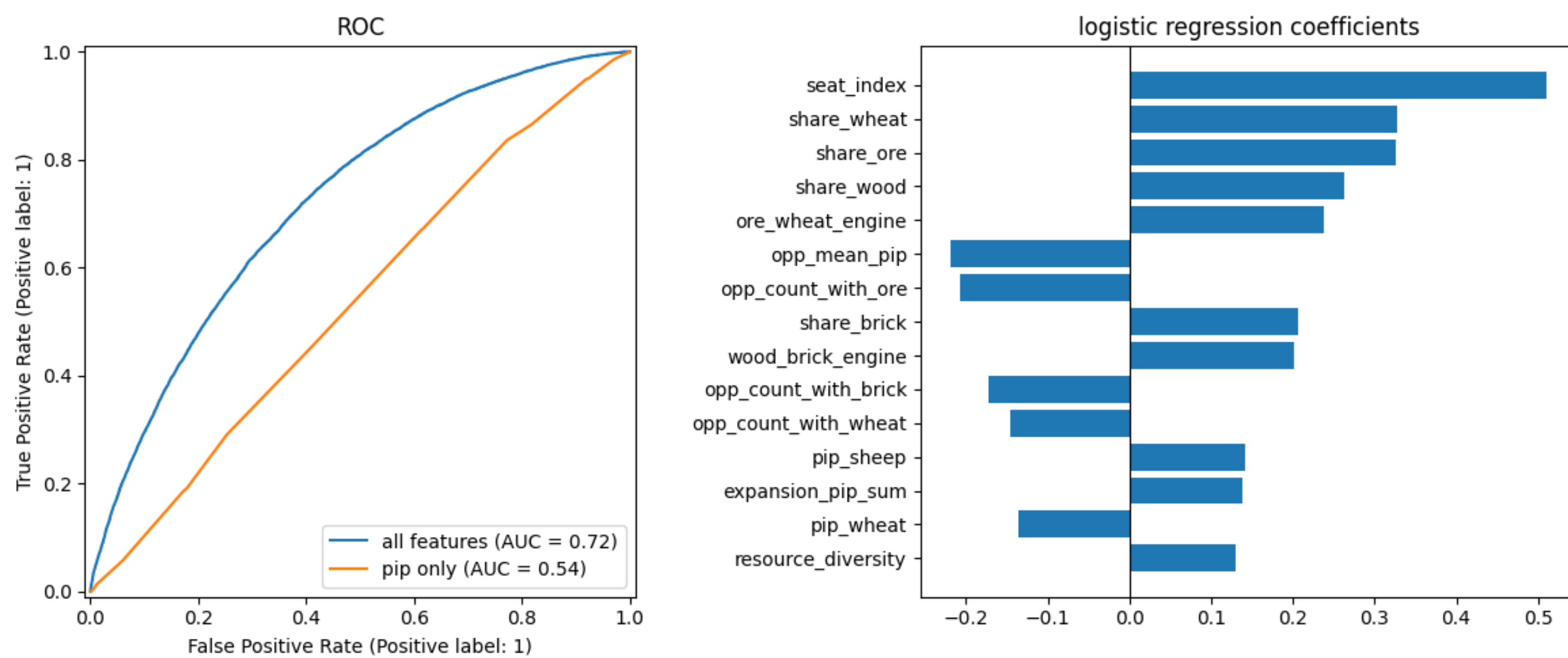


Figure 1. ROC for the full feature set vs. pip-only (left) and the largest logistic-regression coefficients (right).

Opening Archetypes (Unsupervised)

We ran k -means ($k = 5$) on opening features without win labels. The **ore/brick-heavy** archetype (29.4%) is the only one above the 25% base win rate; **sheep-heavy** openings (16.6%) are worst because sheep's main edge — development cards — were rarely bought by the bots, with Largest Army getting claimed in only 0.1% of games.

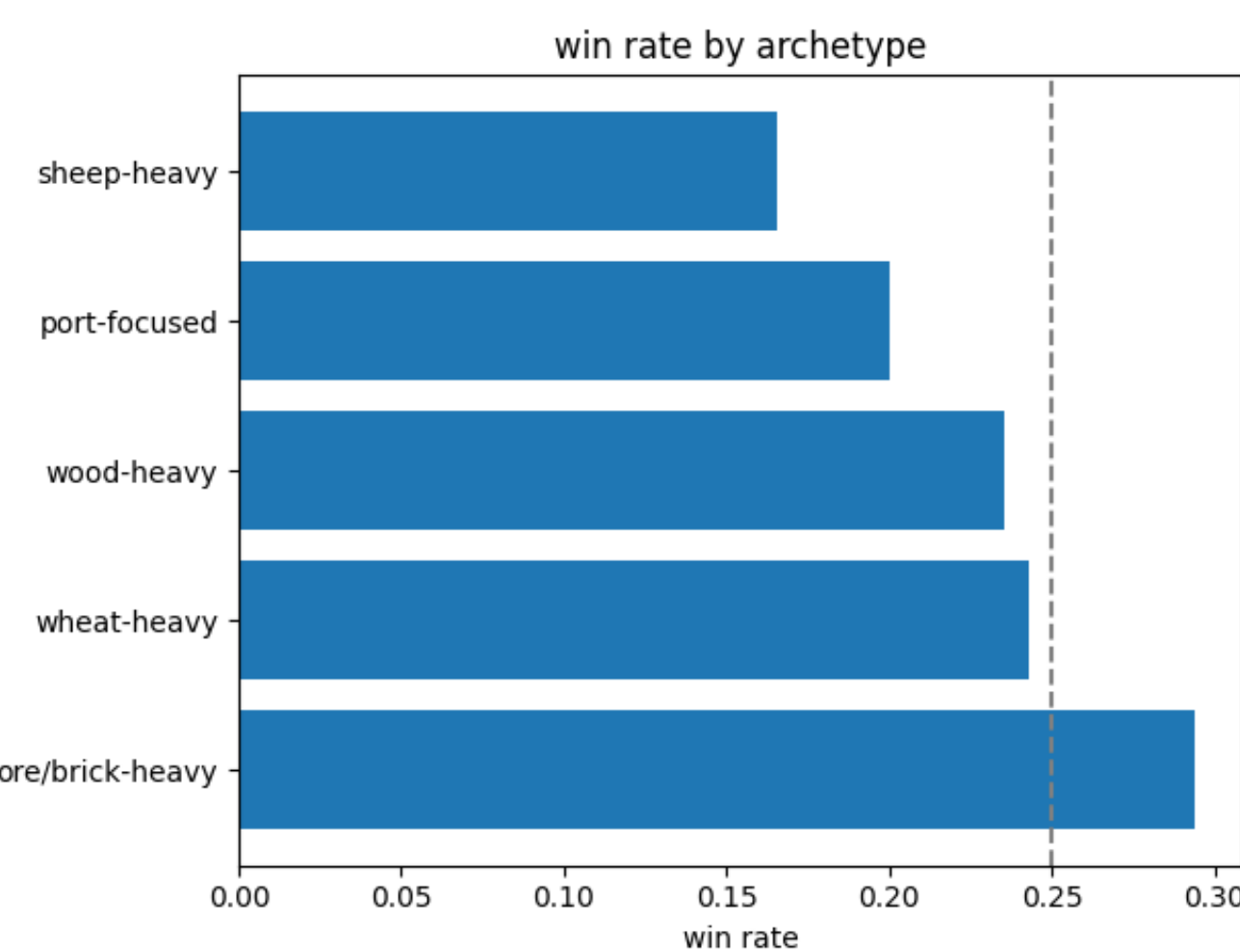


Figure 2. Win rate by unsupervised opening archetype (dashed line = 25% base rate).

Discussion

Pip counting is nearly useless alone (AUC 0.54); a linear model with our features reaches 0.72, and trees and neural nets add almost nothing (0.73). The near-zero gain from nonlinear models is itself a finding: **opening quality is a largely linear, additive function of features** pip counting omits, chiefly resource scarcity and turn order.

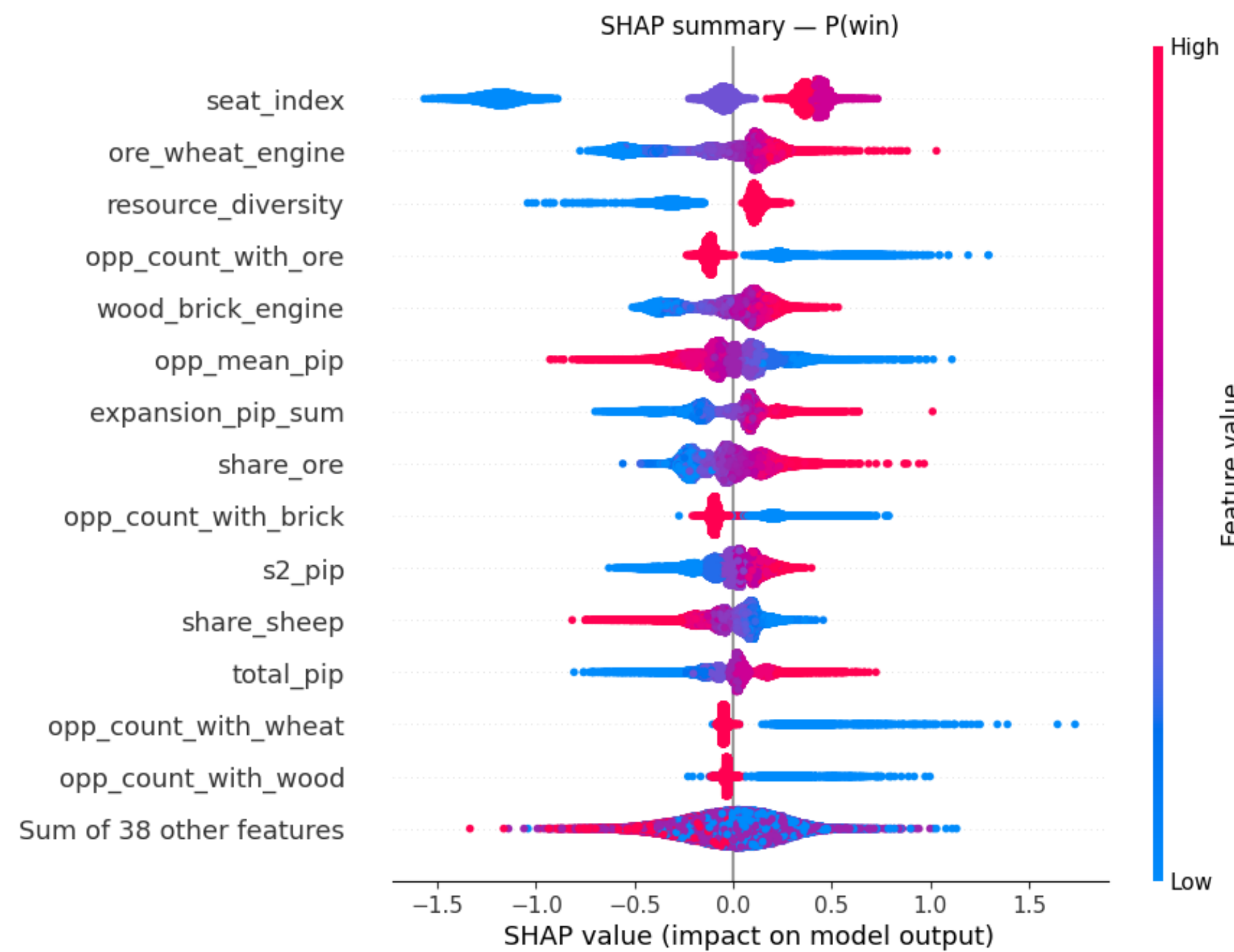


Figure 3. SHAP summary for the XGBoost win classifier; each point is a player-row, colored by feature value.

The RBF kernel underperforming linear models is a clean diagnostic — **the decision boundary is genuinely linear**, not just approximately so. The standout rule is that if a player is not in seat 0 and has both an **ore-wheat** and a **wood-brick engine**, they win **35.9% of the time** (above the 25% base rate), and this rule applies to 46% of all openings. Our regression explains **roughly 20% of the variance in final VP margin from the opening alone** ($R^2 \approx 0.20$), a surprisingly large share for a dice-heavy game and a testament to how much placement matters before a single die is rolled. Five-fold CV intervals are tight (± 0.002 – 0.003) and train/test behavior is matched, indicating minimal overfitting.

Future Work

With more time we would generate data from multiple bot strengths to test whether these signals are agent-specific or intrinsic to Catan, and extend prediction beyond the opening to early-game states. We would also study how optimal placement varies by turn order — since seat index is the single most irreplaceable signal — and test across different player counts, including the popular two-player variant. Finally, we would design opening-aware bots that use our placement signals rather than pip counting and re-run evaluation to measure how much stronger such agents are in practice.

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